CSE4502 (Operating Systems Lab): 2A

Jibon Naher[1] and Samnun Azfar[2]

[1]Assistant Professor, CSE
[2]Junior Lecturer, CSE

February 26, 2026

Part A: Process Scheduling Test Case and CPU Time Slice Analysis

Objective: Demonstrate how different CPU scheduling algorithms allocate CPU time slices to processes.

Setup

Clone the following git repository:

`https://github.com/azfarus/cpu-scheduling-simulation.git`

Git Link for Scheduling Simulation.

Instructions

- Navigate to the `src/` and `include/` directories.
- You must fill up the boilerplate codes in the **Scheduler.hpp** and **Scheduler.cpp** files to implement the scheduling logic for the required algorithms.
- Pay close attention to **process.hpp** and **process.cpp**. These files contain the `Process` class/struct definition, which includes vital attributes such as Arrival Time (AT), Burst Time (BT), program counter, total instructions and Priority. These values are essential for calculating the execution order and priorities in your implementation.

Test Case (5 Processes)

Process	Arrival Time	Burst Time	Priority
P1	0	5	2
P2	1	4	1
P3	2	2	3
P4	3	6	2
P5	4	3	4

Note: Lower priority number means higher priority.

1. FCFS Scheduling Output

Execution Order and Time Slices:

```
=== Gantt Chart ===
+----------+--------+------+-----------+------+
|    P1    |   P2   |  P3  |     P4    |  P5  |
+----------+--------+------+-----------+------+
0          5        9      11          17     20
===================
```

2. SJF (Non-Preemptive) Scheduling Output

Execution Order and Time Slices:

```
=== Gantt Chart ===
+----------+------+------+--------+-----------+
|    P1    |  P3  |  P5  |   P2   |     P4    |
+----------+------+------+--------+-----------+
0          5      7      10       14          20
===================
```

3. Priority Scheduling (Preemptive)

Execution Order and Time Slices:

```
=== Gantt Chart ===
+----------+--------+-----------+------+------+
|    P1    |   P2   |     P4    |  P3  |  P5  |
+----------+--------+-----------+------+------+
0          5        9           15     17     20
===================
```

4. Pre-emptive SJF Scheduling (Shortest Remaining Time First)

Execution Order and Time Slices:

```
=== Gantt Chart ===
+------+------+------+------+--------+-----------+
|  P1  |  P3  |  P1  |  P5  |   P2   |     P4    |
+------+------+------+------+--------+-----------+
0      2      4      7      10       14          20
===================
```

5. Round Robin

Execution Order and Time Slices:

```
=== Gantt Chart ===
+------+------+------+------+------+------+------+------+------+------+------+
|  P1  |  P2  |  P3  |  P1  |  P4  |  P5  |  P2  |  P1  |  P4  |  P5  |  P4  |
+------+------+------+------+------+------+------+------+------+------+------+
0      2      4      6      8      10     12     14     15     17     18     20
===================
```